

ODON INEZA

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EDUCATION

University of Notre Dame

Bachelor of Science in Computer Science; Mathematics Minor

Notre Dame, IN

GPA: 3.73 / 4.00

EXPERIENCE

University of Notre Dame, CVRL

June 2026 – Present

Student Researcher: ArcIris Impostor-Tail Analysis

Notre Dame, IN

- Built a reproducible evaluation pipeline around Notre Dame CVRL's ArcIris matcher (`ndcvr1_002`, ResNet100 + ArcFace; ranked 26th on NIST IREX 10 Identification Track, two-eye by developer, FNIR 0.048 at FPIR = 0.01) that carries a stable `image_id` from ingest through cached embeddings, pairwise scores, and impostor-tail reports
- Extracted iris embeddings once per image (rubber-sheet polar unwrap), reused ArcIris template caches, and sampled up to 100k impostor pairs while keeping every genuine pair—avoiding N^2 model calls on large sets
- Defined the impostor tail as the lowest-score different-eye pairs and wrote an image-level tracker of which irises appear in those pairs, how often, and with whom
- Cross-checked tail hits against ArcIris quality gates (`checkQuality`) to separate segmentation / capture failures from true embedding collisions that justify model changes

University of Notre Dame

May 2026 – June 2026

Research Intern: Face Image Quality CNN

Notre Dame, IN

- Designed and trained a compact SmallResNet (0.33M params) in PyTorch to predict OFIQ face-image quality on 70K FFHQ images, achieving MAE 1.32 and Pearson $r = 0.866$ on a 7K held-out set while cutting overfitting from a $27\times$ to $1.1\times$ val/train MSE ratio vs. ResNet-18
- Built the model as a frozen differentiable quality critic for a GAN generator loss, enabling millisecond OFIQ-like feedback during generative training instead of multi-second full OFIQ scoring

Pivot Access Ltd

Aug 2023 – Jul 2025

Full Stack Developer

Kigali, Rwanda

- Maintained production infrastructure with Kubernetes, Docker, Prometheus, Grafana, Loki, and New Relic, improving observability across multi-service deployments
- Automated CI/CD and monitoring workflows, reducing manual release steps and shortening incident detection time for live services
- Shipped responsive React + Tailwind CSS interfaces that improved usability and engagement across client-facing web applications

PROJECTS

Brain Tumor MRI Classification | *Python, TensorFlow/Keras, Streamlit, scikit-learn*

2026

- Built an end-to-end ML pipeline classifying 7,200 MRI slices into 4 classes (glioma, meningioma, pituitary, no tumor); compared Logistic Regression, scratch CNN, modified CNN, and MobileNetV2
- Achieved 91.4% validation and 86.7% test accuracy with MobileNetV2 transfer learning (vs. 25% majority-class baseline); reduced CNN generalization gap from 9.8% to 2.8% via per-sample normalization

NanoChat (Tiny GPT Playground) | *PyTorch, TypeScript, React, Vite*

2026

- Implemented a trainable GPT-style language model from first principles, covering tokenization, embeddings, attention, and transformer blocks
- Trained models in PyTorch and exported weights for fully in-browser TypeScript inference with a React playground for prompting, sampling, and inspecting generation with no API dependency

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL (PostgreSQL), JavaScript, TypeScript, HTML/CSS

ML / Data: PyTorch, TensorFlow/Keras, scikit-learn, pandas, NumPy, Matplotlib, Plotly, Streamlit

Frameworks: React, Next.js, Flask, Firebase, Redux

Tools: Git, Docker, Kubernetes, Prometheus, Grafana, Loki, New Relic